

Curriculum Vitae



Dr. Bhavesh Kumar Dadhich

Postdoc

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Research Interest

- Non-linear Optics
- Plasmonic nanocrystals
- Femtosecond Laser
- Fluorescence Spectroscopy
- Microscopy

Experience

- Research Experience

- 1. July 2024- Present** Postdoc (PI- Prof. Yonatan Sivan and Prof. Maya Bar-Sadan) at School of Electrical & Computer Engineering and Department of Chemistry, Ben Gurion University of the Negev, Be'er Sheva, Israel-8410501
- 2. Oct. 2021- July 2024** Research Associate-II (PI- Prof. Kamal P. Singh) at Department of Physics, Indian Institute of Science Education and Research Mohali Sahibzada Ajit Singh Nagar, Punjab, India-140306
- 3. Sep. 2014- Dec. 2016** Research Fellow (PI- Prof. Amiya Priyam) at Department of Chemistry, Central University of South Bihar, Gaya, India- 824236

- Teaching Experience

- 1. Nov. 2020- Aug. 2021** Assistant Professor (Physics), Department of Basic Science & Humanities, Aryan Institute of Engineering & Technology, Arya Vihar, Bhubaneswar-752050.

Educational Qualifications

- **2021** **Ph.D. (Physics)**
Department of Physics
School of Applied Sciences
KIIT, deemed to be University, Bhubaneswar – 751024
Thesis title: "**Tunable Plasmonic Silver Nanocrystals: Synthesis, Linear and Non-Linear Optical Properties and Its Applications in Catalysis**"
- **2014** **B.Tech-M.Tech [Integrated]**
Centre for Converging Technologies,
University of Rajasthan, Jaipur, Rajasthan
B.Tech [Nanotechnology] 67.26%.
M.Tech [Nanotechnology] 69.31%.

Accomplishments

- **Senior Research Fellowship (SRF)** is awarded by **CSIR, New Delhi**. CSIR sanction letter no: 09/1035(0014)/2018-EMR-I.
- **International Travel Support (ITS)** by Science and Engineering Research Board (**SERB**), **DST, Govt. of India** to present research work in an international conference “Hot-electron science and microscopic processes in plasmonics and catalysis Faraday Discussion” at **Royal Society of Chemistry, London, United Kingdom** during 18-20 February 2019.
- **Young Scientist Award (YSA)** by Kalinga Institute of Industrial Technology Deemed to be University, Bhubaneswar, Odisha, India.

Publications

(a) Patent

- Kamal P. Singh, Jayant Singh Devra, Sandeep Jakhar, Yateendra Sihag, **Bhavesh K. Dadhich**, Biswajit Panda, “An All-Solid-State High Power Visible Laser System and Method Thereof”, **Application No. 202311016841**.
- Amiya Priyam, **Bhavesh Kumar Dadhich**, and Bhavya Bhushan, “Zero RPM-Synthesized Tunable Plasmonic Hollow Silver Nanoshells in the First and Second Near-IR Windows for Enhanced Solar-Catalysis”, **Application No.201931047638**, The Patent Office Journal No. 51/2019 Dated 20/12/2019.

(b) Journals

- **Bhavesk Kumar Dadhich**, Biswajit Panda, and Kamal P. Singh, “Nano-processing of nanodiamond coated ultrathin glass, semiconductor and metal foils with femtosecond pulses”, [J. Phys. B: At. Mol. Opt. Phys. 58, 025401 \(2025\)](#).
- S Dash, **B K Dadhich**, S K Patra, A Priyam, S Kavita, B Bhushan, “Iron (II)-induced defect luminescence and deconvolution of ferromagnetic and paramagnetic behaviour in $\text{Zn}_{(1-x)}\text{Fe}_x\text{O}$ ($x= 0.01, 0.03, 0.05$) nanoparticles”, [Solid State Communications, 115500 \(2024\)](#).
- **Bhavesk Kumar Dadhich**, Pranati Gupta, Swastik Ballav, Bhavya Bhushan, Prasanta K. Datta, and Amiya Priyam, “No-Stirring Synthesis of sub-50 nm Hollow Silver Nanoshells with Dimethylglyoxime-Induced Plasmons in Visible and Second NIR Windows for Biomedical Applications”, [ACS Applied Nano Materials, 7 \(2023\)](#).
- **Bhavesk K. Dadhich**, Biswajit Panda, Mehra S. Sidhu, Kamal P. Singh, “Nanodiamonds enable femtosecond-processed ultrathin glass as a hybrid quantum sensor”, [Sci Rep 13, 6286 \(2023\)](#).
- **Bhavesk Kumar Dadhich**, Sovan Kumar Patra, Bhavya Bhushan, Ravi Kant Choubey and Amiya Priyam, “Nonlinear Absorption and Refraction of Highly Monodisperse and Luminescent ZnTe Quantum Dots and Its Self-Assembled Nanostructures: Implications for Optoelectronic Devices”, [ACS omega, 2021, 6 \(46\), 31375-31383](#).
- **Bhavesk Kumar Dadhich**, Sayantan Bhattacharya, Swastik Ballav, Bhavya Bhushan, Prasanta K. Datta, and Amiya Priyam, “Femtosecond-Laser-Induced Saturable Absorption and Optical Limiting of Hollow Silver Nanocubes: Implications for Optical Switching and Bioimaging”, [ACS Appl. Nano Mater. 2020, 3 \(11\), 11620–11629](#).
- **Bhavesk Kumar Dadhich**, Bhavya Bhushan, and Amiya Priyam, “Hollow Silver Nanostructures: The Role of Capping Agents in Tailoring the Shape, Structure, and Plasmonic Properties”, [Microscopy and Microanalysis, 2019, 25, 1431-1436](#).
- **Bhavesk Kumar Dadhich**, Bhavya Bhushan, Abhijit Saha, and Amiya Priyam, “Folate-Directed Shape-Transformative Synthesis of Hollow Silver Nanocubes: Plasmon Tunability, Growth Kinetics, and Catalytic Applications”, [ACS Appl. Nano Mater., 2018, 1 \(8\), 4294-4305](#)
- **Bhavesk Kumar Dadhich**, Indrajit Kumar, Ravi Kant Choubey, Bhavya Bhushana, and Amiya Priyam, “Shape and size dependent nonlinear refraction and absorption in citrate-stabilized, near-IR plasmonic silver nanopyrramids” [Photochem. Photobiol. Sci., 2017, 16 \(10\), 1556-1562](#).
- Manmeet Kaur, **Bhavesk Kumar Dadhich**, Ranjit Singh, Kailasa Ganapathi, Toshi Bagwaiya, S. Bhattacharya, A.K. Debnath, K.P. Muthe, S.C. Gadkari, “RF sputtered SnO_2 :

NiO thin films as sub-ppm H₂S sensor operable at room temperature”, [Sensors and Actuators B, 242, 2017, 389–403.](#)

(c) Conference Proceedings

- Subhashree Dash, **BK Dadhich**, A Priyam, SS Meena, S Kavita, B Bhushan, “Structural, magnetic and optical properties of chemically synthesized Zn_{0.98}Fe_{0.02}O dilute magnetic semiconductor nanoparticles”, [Materials Today: Proceedings, 2023, 82, P118-122.](#)
- Javier Aizpurua, Jeremy Baumberg, Alexandra Boltasseva, Phillip Christopher, Emiliano Cortes, Stephen B. Cronin, **Bhaves Kumar Dadhich**, Bart de Nijs, Preeti Deshpande, Yuri Diaz Fernandez, Laura Fabris, Sylwester Gawinkowski, Alexander Govorov, Naomi Halas, Junyang Huang, Bartlomiej Jankiewicz, Rifat Kamarudheen, Jacob Khurgin, Tae Kyung Lee, Julien Mahin, Andrea Marini, Reinhard J. Maurer, Niclas Sven Mueller, Jeong Y. Park, Mahfujur Rahaman, Sebastian Schlücker, Zachary Schultz, Yonatan Sivan, Giulia Tagliabue, Madasamy Thangamuthu, Hongxing Xu and Anatoly Zayats, “New materials for hot electron generation: general discussion”, [Faraday Discuss., 2019, 214, 365-386.](#)
- **B. Dadhich**, S. Kailasa Ganapathi, A. Priyam, Manmeet Kaur, A. K. Debnath, K. P. Muthe, and S. C. Gadkari, “ZnO nanotube array: Gas sensing properties at room temperature”, [AIP Conference Proceedings 1832, 050120 \(2017\).](#)
- **B. Dadhich**, A. Saha and A. Priyam, “ α -hydroxy acids mediated synthesis of hollow silver nanoshells and their optical properties”, [AIP Conference Proceedings 1731, 050072 \(2016\).](#)

References

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- **Prof. Kamal P. Singh**
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